Part A. PERSONAL INFORMATION

CVA date	04/11/2026

First name	Rodrigo		
Family name	Granjel Rodríguez		
Gender	Male	Birth date	09/07/1993
ID number	53817055E		
Email	rodrigo.granjel@bc3research.org	URL Web	https://granjel.github.io/
Open Researcher and Contributor ID (ORCID)		0000-0002-7785-4106	

A.1. Current position

Position	Postdoctoral researcher		
Start date	15/05/2023 (42 months*)		
Institution	Basque Centre for Climate Change (BC3)		
Department	Ecosystems and Global Change		
Country	Spain	Phone number	+34 667 04 86 44
Keywords	Community ecology; Species coexistence; Biotic interactions; Global change; Ecosystem stability and recovery		

*Including 5 months of childcare leave in separate periods between 10/2025 and 09/2026.

A.2. Previous positions

Period	Position / Institution / Country
06/2022 – 03/2023 (10 months)	Fulbright research fellow / North Carolina State University / USA
11/2017 – 03/2022 (53 months)	FPU doctoral researcher / Universidad de Sevilla / Spain
09/2017 – 10/2017 (2 months)	JAE-INTRO research assistant / IRNAS-CSIC / Spain

A.3. Education

Degree / Master's / PhD	University / Country	Year
PhD in Integrative Biology	Universidad de Sevilla / Spain	2023
MSc in Terrestrial Ecology	Universitat Autònoma de Barcelona / Spain	2017
BSc in Biology	Universidad de Vigo / Spain	2016

Part B. CV SUMMARY

Scientific profile and expertise: I am a community ecologist studying how interactions among species and environmental conditions shape biodiversity, ecosystem stability and functioning. I combine experiments with statistical and mathematical approaches to connect observations with ecological theory. My research spans plant communities, plant–herbivore interactions, chemical communication, microbial communities and soil ecology. Across these systems, I ask how coexistence mechanisms arise, how they respond to global change, and what they reveal about the persistence and recovery of ecological communities. I am particularly interested in the relationship between the interaction strengths estimated from data and the mechanisms that permit species to persist. This requires comparing alternative explanations, accounting for uncertainty and being explicit about the scale at which conclusions apply. Rather than treating community patterns as isolated outcomes, I aim to relate them to testable hypotheses about environmental responses, feedbacks and the organisation of ecological interactions.

Research contributions: During my PhD in Integrative Biology at the University of Sevilla, I investigated how nutrient availability and interactions among multiple species affect plant coexistence. My work links nitrogen enrichment and foliar fungal pathogens to structural niche and fitness differences, considering how the number of interacting species changes the importance of direct and indirect effects. I have also studied interaction networks in a multitrophic grassland community, using lasso regularisation and structural stability theory to relate patterns of plant interactions to coexistence mechanisms. These studies connect experimentally measured responses with quantities that help explain community composition.

Complementary research: My work in chemical ecology examines how interactions between plants and their enemies affect communication and resistance. I have investigated volatile-mediated signalling in Arabidopsis thaliana populations with contrasting life histories and contributed to experiments on fungal infection and airborne communication between potato plants. I also collaborate on questions

linking plant diversity to the stability of grassland productivity across spatial scales. Earlier research on peatlands broadened my interests in below-ground communities and carbon cycling. Together, these projects provide complementary ways to study how interactions connect biodiversity with ecosystem processes.

Current programme and international experience: Since May 2023, I have worked as a postdoctoral researcher at the Basque Centre for Climate Change. Within RECODYN, I contribute to experiments, field sampling, data analysis and Bayesian state-space modelling of species interactions, coexistence and community recovery under warming. Through ChEMISt, I participate in experimental and field work connecting phytochemistry, functional traits and multitrophic interactions. My Fulbright research fellowship at North Carolina State University allowed me to investigate the role of interactions in microbial community stability and functioning, hosted by William K. Petry. These experiences have expanded the range of systems and analytical approaches that inform my research.

Research leadership and scientific service: I conceived and led CoVOCom, obtaining AEET funding to investigate coexistence and chemical communication between plants in response to herbivory. I have also contributed as a researcher to collaborative projects led by other investigators, developing experience in both individual and team-based research. My service includes manuscript reviewing, participation in the PCI Ecology recommenders panel, and evaluation of Fulbright predoctoral fellowships and AEET research proposals. I have contributed to scientific meetings through oral presentations, posters and invited seminars, as well as scientific committee work, session coordination and moderation at the 2025 SIBECOL–AEET meeting.

Training, openness and engagement: I have taught undergraduate courses at the University of Sevilla and contributed to postgraduate teaching at the University of the Basque Country. My training activities include co-supervision of an MSc thesis and tutoring research internships through the DAAD RISE Worldwide programme. I use R for statistical analysis, modelling and visualisation, together with version control and reproducible workflows. I share data and analytical code through public repositories, including resources for multispecies coexistence and species-interaction analyses, to make the methods and results accessible for scrutiny and reuse. I have also participated in the European Researchers' Night and school workshops, communicating ecological research to non-specialist audiences. My future research will continue to connect experimental observations, interaction models and biodiversity responses to environmental change.

Part C. RELEVANT MERITS

C.1. Publications

1. *Journal article:* **Granjel RR**, Gross N, Badenhausser I, Arroyo J, Godoy O. 2026. Simple plant–plant interaction patterns predict structural niche and fitness differences in a multitrophic grassland community. OIKOS, e12212 doi: 10.1002/oik.12212

   > Corresponding author and position 1/5 authors. I led data curation, statistical analysis, software development, visualisation and writing, and co-developed the conceptual framework and methods to connect multitrophic interactions with plant coexistence.

2. *Journal article:* **Granjel RR**, Martín-Cacheda L, Röder G, Izquierdo-Ferreiro I, Martín-Díaz A, Picó FX. 2026. Plant–plant signalling reduces herbivore damage in two Arabidopsis thaliana populations with contrasting life histories. ENVIRONMENTAL AND EXPERIMENTAL BOTANY 251, 106472 doi: 10.1016/j.envexpbot.2026.106472

   > Corresponding author and position 1/6 authors. I contributed to study design, experiments, data analysis and writing, and obtained AEET funding for the project on volatile-mediated plant defence in populations with contrasting life histories.

3. *Journal article:* Huang M, **Granjel RR**, Montoya D, Xiao Y, Liu X, Hautier Y. 2026. Plant diversity within communities, not among them, stabilizes grassland productivity across spatial scales. NATURE COMMUNICATIONS 17, 2145 doi: 10.1038/s41467-026-69028-5

   > Position 2/6 authors. I contributed to statistical analyses, interpretation of results and manuscript development in a study combining field surveys and remote sensing across 235 grasslands to assess biodiversity–stability relationships across spatial scales.

4. *Journal article:* Montoya D, Burón-Ugarte A, …, **Granjel RR**, …, Porras-Gómez J. 2025. Complexity and interpretability in global change ecology. PLOS CLIMATE 4, e0000587 doi: 10.1371/journal.pclm.0000587
   > Position 5/10 authors. I contributed to conceptualisation, drafting and revision of a synthesis on complexity and interpretability in global change ecology.
5. *Journal article:* Jiménez-Mejías P, Manzano S, Gowda V, …, **Granjel RR**, …, Fitzpatrick M. 2024. Protecting stable biological nomenclatural systems enables universal communication: A collective international appeal. BIOSCIENCE 74, 467–472 doi: 10.1093/biosci/biae043
   > Position 1280/1563 authors. I endorsed an international appeal for stable biological nomenclature and universal scientific communication.
6. *Journal article:* **Granjel RR**, Allan E, Godoy O. 2023. Nitrogen enrichment and foliar fungal pathogens affect the mechanisms of multispecies plant coexistence. NEW PHYTOLOGIST 237, 2332–2346 doi: 10.1111/nph.18689
   > Corresponding author and position 1/3 authors. I co-designed the research and analyses, performed the research, analysed the data, produced the figures and wrote the manuscript, linking nitrogen enrichment and fungal pathogens to multispecies coexistence mechanisms.
7. *Journal article:* Moreira X, **Granjel RR**, de la Fuente M, …, Rasmann S. 2021. Apparent inhibition of induced plant volatiles by a fungal pathogen prevents airborne communication between potato plants. PLANT, CELL & ENVIRONMENT 44, 1192–1201 doi: 10.1111/pce.13961
   > Position 2/10 authors. I helped design and perform the experiment testing how fungal infection affects volatile-mediated communication between potato plants.
8. *Journal article:* Juan-Ovejero R, **Granjel RR**, Ramil-Rego P, Briones MJI. 2020. The interplay between abiotic factors and below-ground biological interactions regulates carbon exports from peatlands. GEODERMA 368, 114313 doi: 10.1016/j.geoderma.2020.114313
   > Position 2/4 authors. I co-authored a two-year peatland study linking abiotic conditions and interactions among plants, soil fauna and microbes to carbon exports.
9. *Book chapter:* **Granjel RR**, Juan Ovejero R, Briones MJI. 2017. Diversidad edáfica en diferentes hábitats turbosos de la serra do Xistral. CONSERVACIÓN E XESTIÓN DE HUMIDAIS EN GALICIA. HORREUM-IBADER, 31–48. ISBN: 978-84-939661-4-0
   > Position 1/3 authors. I co-authored a chapter on soil biodiversity across different peatland habitats in Serra do Xistral, Galicia.

C.2. Conferences

1. *Invited seminar:* **Granjel RR**. Starting over: Or the hard life of plants in a recovery dynamics experiment. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; June 2026).
2. *Scientific committee member, session coordinator and moderator:* The interplay between species coexistence and abiotic factors: community ecology (general session GSC). III SIBECOL & XVII AEET Meeting (Pontevedra, Spain; June 2025).
3. *Poster presentation:* **Granjel RR**, Montoya D. How does warming affect the coexistence mechanisms of recovering communities? III SIBECOL & XVII AEET meeting (Pontevedra, Spain; June 2025).
4. *Invited seminar:* **Granjel RR**. "`Theory < data`" for community ecology. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; April 2025).
5. *Oral presentation:* **Granjel RR**, Buche L, Spaak J, Godoy O. Environmental variability and the mechanisms behind species persistence. XV AEET National Conference (Plasencia, Spain; October 2021).
6. *Oral presentation:* **Granjel RR**, Allan E, Godoy O. The consequences of nitrogen enrichment and foliar pathogens for multispecies coexistence in grasslands. Festival of Ecology BES (Online due to the COVID-19 pandemic; December 2020).
7. *Poster presentation:* **Granjel RR**. Coexistence and VOC-mediated plant–plant communication in response to insect herbivory (CoVOCom). XVII ECOFLOR Meeting AEET (Bilbao, Spain; March 2020).

8. *Invited seminar:* **Granjel RR**. The struggle for coexistence: Studying biodiversity beyond pairwise interactions. Universität Bern (Bern, Switzerland; October 2019).
9. *Oral presentation:* **Granjel RR**, Gross N, Godoy O. How does the interaction network determine the herbivory effects on plant coexistence? 1st SIBECOL Meeting (Barcelona, Spain; February 2019).
10. *Poster presentation:* **Granjel RR**, Arroyo J, Godoy O. How do biotic interactions and environmental conditions shape species coexistence? Ecology Across Borders 2017 (Ghent, Belgium; December 2017).

C.3. Research projects

1. ChEMISt: Chemical ecology, multitrophic interactions, and stability: unveiling the factors shaping agroecosystem stability under global change: phytochemistry, functional traits, and multitrophic interactions. Ref. 24-C1-Agro-TNA-032. AgroServ Transnational Access, Horizon Europe. Co-PIs: Lucille T. S. Chrétien and Daniel Montoya (BC3); 2024–2026. Awarded access to CNRS-SETE (Moulis, France) and CZU (Prague, Czech Republic) facilities and analytical services for an estimated total value of €85,000.
   > As a beneficiary and postdoctoral researcher, I implement experiments and field sampling linked to RECODYN.
2. RECODYN: Ecosystem recovery dynamics and their response to climate change and habitat fragmentation. Ref. 101043548. ERC Consolidator Grant, European Research Council. PI: Daniel Montoya (BC3); 2023–2027. Total funding: €1,999,315.
   > As a postdoctoral researcher, I conduct experiments, field sampling, data analysis and Bayesian state-space modelling of interactions, coexistence and community recovery under warming.
3. Fulbright Predoctoral Research Fellowship: The role of simple interactions in the stability and function of microbial communities. Fulbright Spain and Regional Government of Andalusia. PI: Rodrigo R. Granjel (Host: North Carolina State University); 2022–2023. Total funding: US$23,660.
   > As the fellowship awardee and project lead, I conceived the study, conducted experiments and modelled microbial interactions, stability and functioning, with William K. Petry as my host.
4. CoVOCom: Coexistence and chemical communication between plants in response to insect herbivory. Early-career ecology project grants (“Tomando la iniciativa”), Spanish Association of Terrestrial Ecology (AEET); 2020. Total funding: €2,000.
   > As principal investigator, I conceived the project, secured funding, and coordinated and conducted experiments.
5. MeDiNaS: Mecanismos de mantenimiento de biodiversidad en ambientes variables. Ref. RTI2018-098888-A-I00. Plan Nacional de I+D+i / Retos Investigación, Spanish Ministry of Science, Innovation and Universities. PI: Oscar Godoy (University of Cádiz); 2019–2021. Total funding: €105,270.
   > I contributed as a work-team member and predoctoral researcher for 28 months.

C.4. Knowledge and technology transfer

I contribute to open and reproducible ecology by sharing research data and analytical code in public repositories. My contributions include an MIT-licensed software release for estimating species interactions using lasso regularisation and calculating structural niche and fitness differences (2026; doi: 10.5281/zenodo.23054165), and a PaNDiv data-and-code archive released under CC BY 4.0 (2022; doi: 10.5281/zenodo.7343389). By making these analytical workflows available alongside the underlying data, I help other researchers reproduce the analyses and adapt the methods to their own ecological questions.

I have also participated in public science outreach through the European Researchers’ Night and school workshops, sharing ecological research with non-specialist and younger audiences. These activities complement the dissemination of my research within the scientific community.